

Sales Forecasting Using Machine Learning

EncoderX Data Science Internship – Batch 02

Task 2: Sales Forecasting

1. Introduction

Sales forecasting helps businesses estimate future demand using historical sales data. In this project, machine learning techniques were used to analyze sales patterns and predict future sales.

The project focuses on data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and business insights.

2. Dataset

The dataset contains daily sales information for **10 stores** covering **2022–2023**.

Main features include:

- Date
- Store
- Sales
- Promotion
- Holiday

The dataset was cleaned and organized chronologically before model development.

3. Data Preprocessing

The following preprocessing steps were performed:

- Checked and handled missing values.
- Checked for duplicate and invalid records.
- Converted the date column into datetime format.
- Checked for negative sales and invalid values.

- Organized the data chronologically.

Outlier Handling

The IQR method identified **54 potential outliers (1.1%)** in the training data.

These values were reviewed but retained because they may represent legitimate sales variations caused by holidays, promotions, or other business factors.

4. Exploratory Data Analysis

EDA was performed to understand the main sales patterns.

The analysis examined:

- Yearly sales trends
- Monthly sales trends
- Store performance
- Promotion effects
- Weekday and weekend sales
- Holiday effects
- Promotion trends over time
- Relationships between sales, promotions, and holidays

The EDA helped identify useful patterns for feature engineering and forecasting.

5. Feature Engineering

Several time-based and historical sales features were created:

- Year
- Month
- Day
- Quarter
- Day of week
- Weekend indicator
- Season
- Promotion
- Holiday

- 1-day lag
- 7-day lag
- 30-day lag
- 7-day rolling average
- 30-day rolling average

Lag and rolling features were created using previous sales values to capture recent sales trends.

6. Train, Validation and Test Split

The data was divided chronologically to avoid using future information for training.

Dataset	Rows
Training	4,900
Validation	1,050
Test	1,050

The split was approximately:

- 70% Training
- 15% Validation
- 15% Test

7. Model Development

Two machine learning models were developed:

1. **Linear Regression**
2. **Random Forest Regressor**

The models were trained using the engineered features.

8. Model Evaluation

The models were evaluated using:

- Mean Absolute Error (MAE)

- Root Mean Squared Error (RMSE)
- R^2 Score

Validation Results

Model	MAE	RMSE	R^2
Linear Regression	7.25	8.95	0.867
Random Forest	5.54	6.97	0.920

Random Forest performed better than Linear Regression on the validation data.

Final Test Result

The final Random Forest model achieved:

- **MAE: 6.63**
- **R^2 : 0.877**

This indicates that the model was able to produce reasonably accurate predictions on unseen test data.

9. Forecast Results

The final Random Forest predictions were compared with actual sales.

Test Period Summary

- Average actual sales: **243.54**
- Average predicted sales: **239.66**
- Total actual sales: **255,717.30**
- Total predicted sales: **251,638.04**

The model slightly underestimated total sales but followed the overall sales pattern reasonably well.

The actual vs predicted and overall forecast charts are included in the project files.

10. Feature Importance

The most important features identified by the Random Forest model were:

Feature	Importance
Day of week	27.5%
Holiday	23.3%
Promotion	22.2%
30-day rolling average	20.3%

These results show that calendar patterns, holidays, promotions, and recent sales trends have a strong influence on sales predictions.

11. Business Insights

- **Promotions boost sales:** Average sales with promotions were higher than sales without promotions.
- **Holidays increase sales:** Holiday periods showed higher average sales.
- **Weekdays perform better:** Average weekday sales were higher than weekend sales.
- **Day of week was the strongest predictor:** It had approximately 27.5% feature importance.
- **Recent trends matter:** The 30-day rolling average was also highly important.
- Store differences were relatively small compared with calendar and promotional factors.

12. Recommendations

- Plan promotions around lower-sales days to improve demand.
- Prepare sufficient inventory for holidays and high-demand periods.
- Monitor recent sales trends to identify changes in demand early.
- Use sales forecasts for inventory, marketing, and staffing decisions.
- Add product, price, and customer-level data in future versions to improve forecasting.

13. Conclusion

This project developed a sales forecasting solution using historical store sales data and machine learning.

Two models were compared, and **Random Forest performed better than Linear Regression**. The results showed that day of week, holidays, promotions, and recent sales trends were important factors for predicting sales.

Overall, the forecasting model can help businesses make better decisions related to **inventory, marketing, staffing, and sales planning**.

14. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Joblib
- Google Colab

15. Project Deliverables

- Jupyter Notebook
- Trained model files
- Processed dataset
- Forecast results CSV
- Forecast visualizations
- Project report
- GitHub repository